OT Pentesting Source Map

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Summary

paulveillard/cybersecurity-OT links the most literature (~170 links), followed by the jassics IoT/hardware list (~100) and the jassics ICS/OT list (~75). TeamStarWolf is the opposite: few links, but the most original written content (the ICS/OT Reference is a 54 KB field guide with protocol tables, malware deep dives and checklists).

Source	Content links (approx.)	Character
paulveillard/cybersecurity-OT	~170	Broadest OT link hub: papers, PCAPs, emulators, OSINT engines, news; some dead links
jassics IoT/hardware list	~100	Books, firmware/radio tooling, labs, certs
jassics ICS/OT list	~75	Tight learning path: books, courses, labs, tools, certs
TeamStarWolf ICS/OT discipline page	~35	Learning path + curated tools, mapped to ATT&CK and 800-53
OWASP Agentic Skills Top 10	~25 references	2026 incident timeline and research list for AI-skill security
TeamStarWolf ICS/OT Reference	~15	Content, not links: Stuxnet/TRITON/Industroyer, attack methodology, IR

The map below sorts your 32 links into six categories, then categorizes everything those hubs link to, deduplicated, one level deep (see Notes for where the crawl stops).

Your sources, by category

Your 32 links fall into six categories; two are duplicates, two are outdated versions, and one (BypassPro) is web-only, not OT.

1. Standards and frameworks

Source	What it is	Note
NIST SP 800-82r2 PDF	Guide to ICS Security (2015)	Superseded by r3 (Sep 2023)
NIST OT Security project	NIST OT program hub	You listed it twice. "30 November" is the comment deadline for the SP 800-82r4 initial public draft, not a final release
MITRE ATT&CK for ICS matrix / techniques	Adversary TTP framework for ICS	Same framework, two views; TeamStarWolf tracks deprecations (ATT&CK v19.2)
OWASP OT Top 10 2025	Top 10 OT risks (unknown assets, known-vuln devices, supply chain, availability, access control, detection, zones/conduits, awareness, weak devices, hardening)	Has a standards mapping table and PDF
OWASP WSTG v4 PDF	Web testing guide	Archived v4; current is the online WSTG
OWASP ISTG / GitHub	IoT Security Testing Guide	Same project, two links
OWASP ISVS	IoT security verification requirements	Pairs with ISTG (requirements vs. test cases)
OWASP Agentic Skills Top 10	AST01–AST10 risks for AI agent skills	v1 2026, incubator; directly relevant if you build pentest skills

2. OT pentest methodology and protocol primers

Source	What it is	Note
ACI Europe airport OT/ICS pentest guidelines	Sector guideline (airports)	Only sector-specific source in your set
InfoSec Write-ups: OT pentest methodology	"Test without breaking it" methodology article	Site blocked automated reading; links not crawled

Source	What it is	Note
Aravind: attacker's mindset, part 4	Intro OT pentest series	Medium blocked reading; parts 1–3 likely worth reading too
Frenos OT pentest checklist	10-step before/during/after checklist	Vendor content: solid safety steps, but cost and coverage figures push Frenos' digital-twin product
CSI Modbus 101	Modbus primer	Protocol basics before testing

3. Training and video

Source	What it is
YouTube talk	Single OT video (not opened)
UtilSec channel	Utility/OT security channel
HTB Academy paths	Structured pentest paths; HTB also has an ICS segmentation post

4. Lab

Source	What it is
GRFICSv3	Dockerized chemical-plant lab: OpenPLC, ScadaLTS HMI, engineering workstation, firewall, Kali, MITRE Caldera (OT plugin), optional Wazuh + Zeek; 278 stars, GPL-3.0

5. Curated lists and references (the link hubs)

Source	Focus
paulveillard/cybersecurity-OT	OT link hub
jassics ICS/OT / IoT / repo root	Learning lists per discipline
TeamStarWolf discipline page / Reference	Learning path + field guide

6. AI-assisted pentesting and Claude

Source	What it is	Note
0xSteph/pentest-ai	AI pentester (MCP/CLI/CI) that re-runs exploits via "oracles" and ships replayable proof capsules; reports + SARIF	1.7k stars, MIT; web/bug-bounty scope, no OT
frendysanusi/claude-pentest-skills	Claude Code skill pack: /scope, /recon, /hunt, /validate (6 gates), /report to PDF; OWASP WSTG-mapped	24 stars; web-only, good template for an OT skill
nbshenxm/pentest-agent	Research framework: recon, planning, execution agents; benchmarked on Vulhub	Backed by an ACM paper (arXiv 2411.05185)
0x727/BypassPro	Burp extension for 401/403 and WAF bypass	Not AI, not OT; web tooling
anthropics/skills	Official Anthropic skill examples	Reference format for writing your own skills
Claude Security tutorial	Anthropic Academy intro to Claude Security	Codebase vuln finding and patching, not network/OT testing

Linked literature: ICS/OT

Deduplicated across the three OT hubs. **From:** J = jassics ICS/OT, P = paulveillard, T = TeamStarWolf. Items in two or more hubs are the consensus picks.

Books and long-form

Resource	Note	From
Hacking Exposed: ICS (Bodungen et al.)	Foundational ICS pentest book	J, T
Pentesting Industrial Control Systems (Paul Smith)	Hands-on lab-based pentesting	J
Industrial Cybersecurity, 2nd ed. (Ackerman)	Monitoring and architecture; excerpt PDF	J, P, T

Resource	Note	From
Industrial Network Security, 2nd ed. (Knapp, Langill)	Classic reference; P rates it best	J, P
Countdown to Zero Day (Zetter)	Stuxnet narrative	J, T
Practical Industrial Cybersecurity (Brooks, Craig)	ICS security fundamentals	J
ICS Cybersecurity Field Manual (Ackerman, Dragos)	Free, 3 volumes	J
The ICS Cyber Kill Chain (Assante, Lee)	Two-stage ICS attack model	J, P
Navigating Industrial Cybersecurity field guide	Basic field guide	P
SCADA for Relay Technicians	SCADA beginner slides	P
Cybersécurité des systèmes industriels (Flaus)	French-language book	P

Standards, guidance and frameworks

Resource	Note	From
NIST SP 800-82r3	Current OT guide (2023); r4 in draft	J, P, T
ISA/IEC 62443 series	Zones/conduits, security levels	J, T
NERC CIP	North American grid, mandatory	T
NIST CSF 2.0	General framework	P
DoD Control Systems SRG	Requirements guide	P
BSI OPC UA Security Analysis	German gov protocol analysis	P
BSI ICS Top 10 Threats	Predecessor idea to OWASP OT Top 10	P
CWE view 1424: weaknesses vs. 62443 / CWE ICS/OT SIG	Weakness taxonomy for findings	P
IACS STAR calculator	OT vuln rating method	P
Water sector IR guide (IC3)	Sector incident response	P

Resource	Note	From
EBIOS RM / COMAH / NIS per EU state	FR risk method, UK hazard regs, EU NIS	P
ENISA SCADA	EU training material	J

Courses and certifications

Resource	Note	From
CISA ICS training	Free, 100–400 series	J, P, T
CISA NICCS catalog	Free federal courses	T
Dragos learning	Free content	J
SANS ICS410 / ICS515 / ICS612 / ICS456	Paid; map to GICSP / GRID / GCIP	J
GICSP / GRID / GCIP	GIAC ICS certs	J, T
ISA/IEC 62443 certificate program	Tiered standard certs	J, T
Claroty Academy	Vendor training	J
ICS 301v review	CISA course review	P
BHIS webcasts / SANS ICS resources	Free webcasts, posters	T

Labs, simulators and CTFs

Resource	Note	From
GRFICSv2	Predecessor of your GRFICSv3	J
OpenPLC / Docker image	Open-source PLC for home labs	J, P, T
ControlThings	Linux distro with ICS tools	P, T
FUXA / Rapid SCADA / ScadaBR	Open-source HMI/SCADA	P
NetToPLCSim / LogixPro / dnp3slavesim / snap7 Docker	Siemens, Allen-Bradley, DNP3, S7 simulators	P
Conpot / GasPot / HoneyPLC / T-Pot	ICS honeypots	J, P

Resource	Note	From
SANS/Dragos ICS CTF / Splunk BOTS Dragos / NPS Labtainers	Free challenges	P
GRFICSv2 hacking part 1 / part 2	Walkthroughs, mostly apply to v3	P
OpenPLC + Factory I/O / Cantera virtual lab / biero S7 lab	Lab build tutorials	P
Fortiphyd YouTube	GRFICS authors' channel	P

Tools

Resource	Use	From
Redpoint / s7scan / PLCScan	ICS enumeration (use cautiously on live OT)	J, T
GRASSMARLIN	Passive topology mapping (archived)	T
ISF / Metasploit	Exploitation frameworks	J
pymodbus / opendnp3 / libiec61850 / Scapy / Wireshark	Protocol stacks and analysis	J, T
Zeek / ICSPcapViz	Passive monitoring, PCAP visualization	P, T
MITRE caldera-ot	OT adversary emulation (in GRFICSv3)	T
ICSpector	Microsoft PLC forensics	P
binwalk / FACT / EMBA	Firmware analysis	T
Dragos, Claroty, Nozomi, Armis, Tenable OT, Cisco Cyber Vision	Commercial passive monitoring	J, T

Datasets and PCAPs

Resource	Note	From
ICS-pcap	Multi-protocol captures	T
s7-pcaps	S7-300/400 traffic	P
ICS Defense PCAPs	PCAP archive	P

Resource	Note	From
Electra dataset / OPC UA dataset	Anomaly-detection datasets	P
CloudShark Modbus	Online capture	P

Research papers, talks and write-ups

Resource	Topic	From
The Spear to Break S7CommPlus	Siemens protocol attack (BH EU 2017)	P
Web-based PLC malware	NDSS 2024	P
PLC vulns, attacks, detection, forensics survey	MDPI survey	P
Evil PLC attack / OPC UA attack surface / Fuxnet	Claroty Team82 research	P
Sandworm OT attack 2022 / Pipedream	Incident analyses	J, P
Forescout: Denmark/Ukraine energy attacks	Incident analysis	P
AMNESIA:33	TCP/IP stack vulns	P
Pwn2Own OPC UA bypass	CVE-2022-29865	P
UMAS protocol reversing	Schneider protocol	P
DEF CON 26: Hacking PLCs	OpenPLC talk	P
Virtual PLC platform thesis / ICS honeypots-in-a-box	Academic	P

Threat intel, news and OSINT

Resource	Note	From
Dragos Year in Review / threat groups	Annual report, actor tracking	J, T
CISA ICS advisories	Primary vuln advisories	J
Claroty Team82 / Nozomi blog	Vendor research	J
Siemens / ABB / Schneider / Dragos	Vendor PSIRTs	P

Resource	Note	From
SecurityWeek ICS / Industrial Cyber / SCADAhacker	News and blogs	J, P
S4 talks / ICS Village / SANS ICS Summit	Conference video	J, T
Shodan / Censys / ZoomEye / FOFA / Onyphe / Shadowserver map	Exposure search (research only)	P
Kunyu / grinder / ICSRank	OSINT automation	P

Linked literature: IoT and hardware

All from the jassics IoT/hardware list, plus paulveillard's small hardware section (P). This is the bridge to PLC firmware and field-device work.

Books

Resource	Note
Practical IoT Hacking (Chantzis et al.)	Current standard text
The Hardware Hacking Handbook (van Woudenberg, O'Flynn)	Fault injection, side-channel
Attacking Network Protocols (Forshaw)	Proprietary protocol RE; applies to ICS protocols
The IoT Hacker's Handbook (Gupta)	IoT exploitation
The Car Hacker's Handbook (Smith)	CAN bus; free
Practical Hardware Pentesting (Valle)	UART/JTAG/SPI workflow
Embedded Systems Security (Kleidermacher)	Design-side view

Methodologies and standards

Resource	Note
OWASP FSTM	Firmware security testing methodology
OWASP ISVS / IoT Top 10	Already in your set (ISVS)

Resource	Note
MITRE EMB3D	Embedded-device threat model (2024); TeamStarWolf has an EMB3D reference
NISTIR 8259 / 8228	IoT device cybersecurity baselines

Courses, labs and certs

Resource	Note
Microcorruption	Free MSP430 embedded exploitation CTF
Azeria Labs ARM	Free ARM assembly and exploitation
OWASP IoTGoat / DVRF	Vulnerable firmware to practice on
FirmAE / Firmadyne	Firmware emulation
SANS SEC556 / Attify OIP / 8kSec / OffSec EXP-401	Paid training and certs
Make Me Hack	Hardware RE video series (P)

Tools

Area	Tools
Firmware / RE	binwalk, FACT, EMBA, Ghidra, radare2, Qiling, Unicorn, LIEF, trufflehog
Hardware interfaces	Bus Pirate, JTAGulator, Saleae, ChipWhisperer, J-Link
Radio / SDR	HackRF, RTL-SDR, GNU Radio, URH, KillerBee, Proxmark3, Flipper Zero
IoT protocols	MQTT Explorer, libcoap, modbus-cli
Datasheets (P)	alldatasheet, datasheetcatalog, Falstad circuit sim

Blogs and talks

Pen Test Partners, NCC Group hardware research, Quarkslab, Attify blog, IoT Village, Hardwear.io talks, Joe Grand, stacksmashing.

Linked literature: AI pentesting and agent skills

None of the AI tools you collected target OT; all are web/network-focused. The closest path to an OT skill is claude-pentest-skills' structure (scope gate, per-class methodology, validation gates, report) applied to the OT sources above, hardened against the OWASP AST10 risks.

Linked from the AI tools

Resource	What it adds	From
PentestAgent paper (arXiv 2411.05185) / ACM DOI	Recon, planning, execution agent design with RAG over CVE/exploit sources	pentest-agent
Vulhub / cvemap	Benchmark targets; CVE lookup	pentest-agent
Why verification	Design rationale: model proposes, oracle verifies	pentest-ai
Benchmarks / Juice Shop methodology	How results were measured	pentest-ai
pentest-ai-agents	Claude Code subagent files	pentest-ai
CI/CD guide / bug-bounty guide	SARIF export, scoped submissions	pentest-ai
OWASP WSTG	Methodology the skills map to	claude-pentest-skills
AutoBypass403	Upstream of BypassPro	BypassPro

Linked from OWASP Agentic Skills Top 10

Resource	What it adds
AST10 whitepaper v1 (PDF)	Full risk write-ups
Security assessment checklist	Review installed or self-written skills
Skill development guide	Least-privilege manifests, pinning, signing
Universal skill format	Proposed cross-platform manifest with permission allowlists
Incident response playbook / case studies	Skill compromise handling

AST10 also cites research by name without links: Snyk ToxicSkills (Feb 2026), Check Point on Claude Code CVE-2025-59536 and CVE-2026-21852, Trail of Bits on skill distribution (Jun 2026), Air Security's SkillJacking (Jul 2026), and arXiv 2601.17548 and 2602.06547 (USENIX Security 2026). Relevant if your OT skill pulls instructions or payloads from external URLs (AST05).

Notes and crawl limits

The crawl went one level deep; the hubs below are themselves link lists, so "out of links" was reached only for the pages actually read.

Next-layer hubs (not crawled): hslatman/awesome-industrial-control-system-security, Orange-Cyberdefense/awesome-industrial-protocols (offensive, per protocol; the best next read for pentesting), ITI/ICS-Security-Tools, SCADAhacker library, ICSCSI library, Robert M. Lee's starter collection, MrM8BRH ICS list, fkie-cad/awesome-embedded-and-iot-security, awesome-hardware-hacking, and TeamStarWolf's ICS ATT&CK Atlas and Firmware & IoT Reference.

Could not be read: the InfoSec Write-ups article and the Medium article (bot blocking), and the jassics repo root (robots.txt). Their linked resources are not in this map.

Left out on purpose:

- Entries paulveillard marks as down: Principles of Information Security PDF, INL CCE reference document, GreHack code-injection paper, Redfox PLC hacking post.
- paulveillard's "Biero resource list" (a joke link), its job board, and a Google Drive copy of Industrial Network Security (publisher link used instead).
- Generic vendor home pages and certifications only loosely tied to OT (ISACA, CREST, EC-Council).

Overlap to note: jassics ICS and TeamStarWolf largely agree (CISA training, SANS/GIAC, 62443, ATT&CK, Dragos YIR, the same 3 books). paulveillard adds what the others lack: protocol papers, PCAPs, emulators, OSINT engines, non-English sources.

Sources

Pages opened for this map: paulveillard/cybersecurity-OT · jassics ICS/OT · jassics IoT · TeamStarWolf discipline · TeamStarWolf Reference · NIST OT project · OWASP OT Top 10 · OWASP AST10 · GRFICSv3 · Frenos checklist · pentest-ai · claude-pentest-skills · pentest-agent · BypassPro. Other entries in "Your sources" are described from their URL and publisher, not opened.